

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	I	{Maintain/ control/ prevent changes in} pH (1) keeping pH at optimum/ which would alter the {rate of reaction/ time taken}/ which could denature the enzymes/ prevents coagulation of milk (1)	1	1		2		2
			II	{Maintain/ control/ prevent changes in} temperature/ acclimatise to the same temperature (1) so that all of the enzymes and substrates had the same <u>kinetic</u> energy / if some test tubes were incubated at a higher temperature the enzymes (and substrates) would have more <u>kinetic</u> energy / ORA (1) Ignore references to denaturing/ optimum	1	1		2		2
		(ii)		91.2		1		1	1	1
		(iii)		Correct plots (2) $\pm \frac{1}{2}$ small square 5 correct = 2 marks 4 correct = 1 mark 0/1/2/3 = 0 marks Centre of plots joined with a ruler, no extrapolation/ line of best fit with ruler (allow extrapolation) (1) Range bars correct (1) $\pm \frac{1}{2}$ small square		4		4	4	4

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	(b)	(i)		X = 1.6 - 1.8% <u>and</u> Y = 4.3 – 4.5% Take from candidates graph		1		1	1	1
		(ii)		Any two (x1) from: • Comparison of {range bars/ data} for {1%/2%/ X/ lower concentrations} (smaller) compared to {3%/4%/ 5%/ Y/ higher concentrations} (larger) (1) • Comparison of degree of overlap of the range bars e.g. at 3%, 4% and 5% all overlap and no overlap between 1 and 2% (1) • Correct reference to {reliability/ repeatability/ consistency} linked to {length of range bar/ spread of data} OR there would be less confidence in the trend for the overlapping results. (1) Reject reference to accuracy/ reproducibility			2	2		2
		(iii)		{Judging/ subjectivity of} {when the solution becomes transparent/ the end point } (1) As the milk powder forms a suspension, the powder may form sediment at the bottom off the test tube altering the time taken for the milk to clear / not mixed evenly/ not all transferred from one test tube to another (1)			2	2		2
				Question 3 total	2	8	4	14	7	14